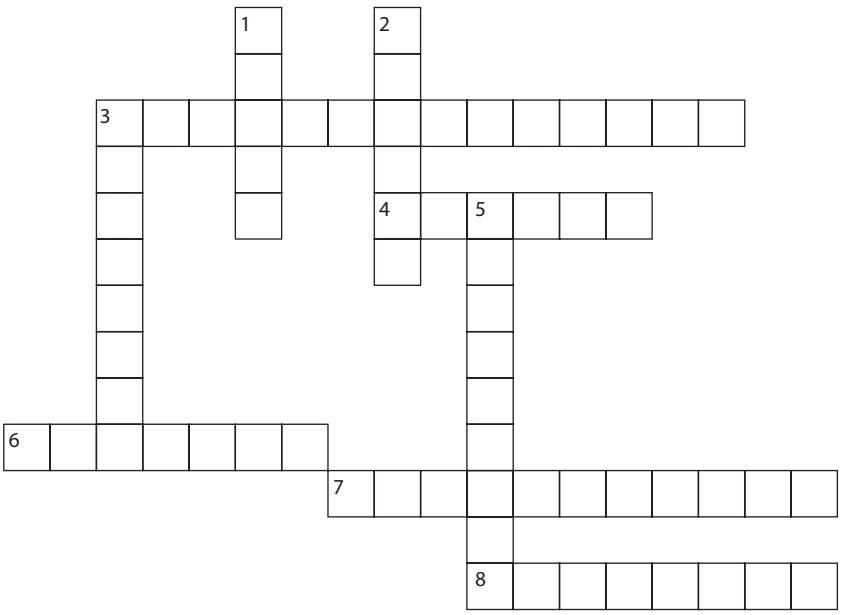
Chemical reactions in nature

Two chemical reactions that occur in nature are **photosynthesis** and **respiration**. Photosynthesis is the process by which green plants use carbon dioxide and water to produce food (glucose) and oxygen. Respiration is the process by which living things use food (glucose) and oxygen to produce energy. Carbon dioxide and water are also produced. All living things respire. The word equations for these reactions are:

Photosynthesis: carbon dioxide + water ⟶ glucose + oxygen

Respiration: glucose + oxygen ⟶ carbon dioxide + water + energy
released

☞ **1** Use the words in the word bank to complete the crossword puzzle.



Word bank

respire
oxygen
equations
photosynthesis
sunlight
water
energy
chlorophyll
products

Across

3 The process by which green plants make food

4 Released when respiration occurs

6 All living things do this

7 Found in the leaves of green plants

8 Needed for photosynthesis to occur

Down

1 One of the reactants of photosynthesis

2 Glucose and _____ are the reactants for respiration.

3 Glucose and water are the _____ of photosynthesis.

5 Chemical reactions can be written as word _____.

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